

STUDY NOTES — CHAPTERS 5, 6 & 7

Thorax & Rib Cage · Skull Morphology & Sexual Dimorphism · Age Estimation

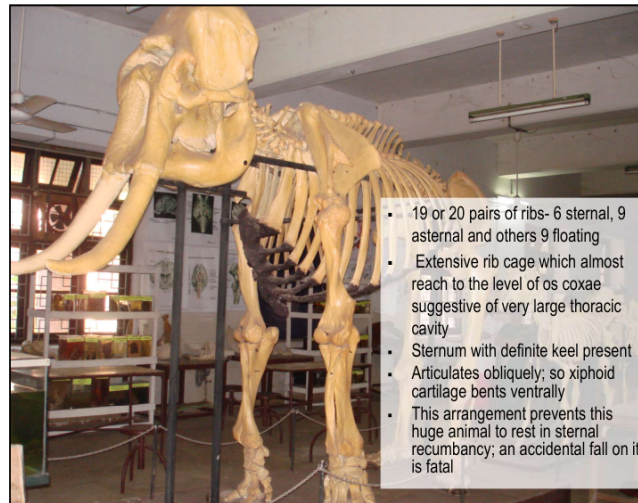
Biological and Anatomical Aspects of Elephants

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CONTENTS: Ch.5 — Thorax & Rib Cage | Ch.6 — Skull Morphology & Sexual Dimorphism | Ch.7 — Age Estimation

5. Thorax & Rib Cage



10

Slide 10 — Thoracic skeleton: extensive rib cage reaching to os coxae level

5.1 Rib Count & Structure

Parameter	Asian Elephant	African Elephant
Total rib pairs	19–20	21
Sternal ribs (direct sternum attachment)	6	6
Asternal ribs (costal cartilage)	9	9
Floating ribs (free)	~4–5	~6
Rib cage extent	Reaches nearly to os coxae level	Similar
Thoracic cavity	Very large — accommodates 12–21 kg heart	Very large

- Sternum: definite keel (carina) present — unlike flat sternum of most domestic animals
- Xiphoid cartilage: articulates obliquely, bends ventrally — relevant to anaesthesia positioning
- Costal cartilages: wide; calcify with age — useful at necropsy for gross age estimation

5.2 No Pleural Space — Unique Respiratory Anatomy

Elephants are the **ONLY** mammals with no true pleural space. Both visceral and parietal pleura are composed of dense connective tissue and are **FUSED** together. Lungs are physically adhered to the chest wall. Breathing is driven **ENTIRELY** by diaphragm movement.

Species	Rib Pairs	Pleural Space	Breathing Mechanism
Asian Elephant	19–20	ABSENT — pleura fused	Diaphragm ONLY
African Elephant	21	ABSENT — pleura fused	Diaphragm ONLY
Horse	18	Present — fluid-filled	Diaphragm + rib cage expansion
Cattle	13	Present — fluid-filled	Diaphragm + rib cage expansion
Dog	13	Present — fluid-filled	Diaphragm + rib cage expansion

- Diaphragm: unusually thick and muscular; generates all inspiratory pressure alone
- Resting respiratory rate: ~4–8 breaths/min; large tidal volume compensates
- Evolutionary reason: semi-aquatic ancestry — swimming elephants breathe via trunk-snorkel

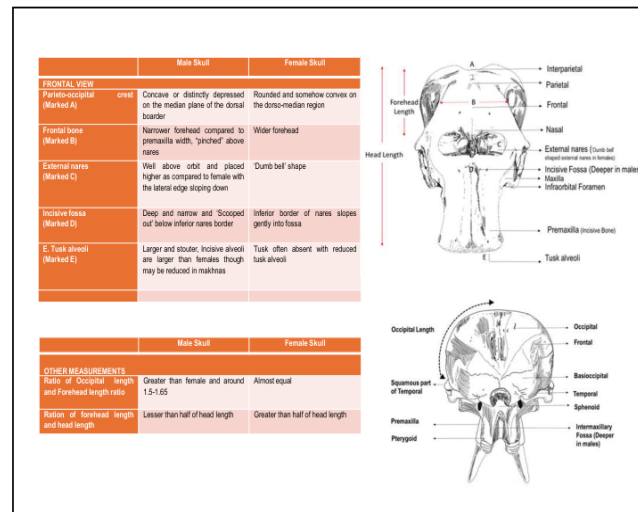
- West JB (2001): confirmed biophysical significance for deep-water diving/swimming

5.3 Necropsy Findings — Thorax

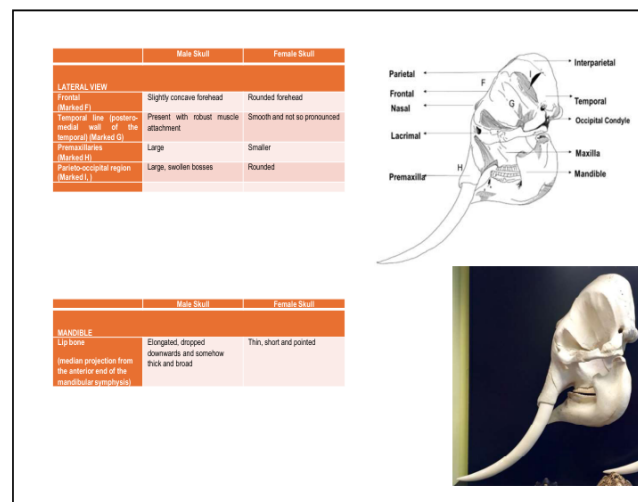
- Lungs appear FIRMLY ATTACHED to inner chest wall — NORMAL anatomy, not adhesion from pleuropneumonia
- Any fluid between chest wall and lung surface is PATHOLOGICAL (haemorrhage, exudate, oedema)
- Heart: centrally located; BIFID (double-pointed) apex — distinctive and normal; weight 12–21 kg
- Trachea: large-diameter; check for foreign bodies, mucus plugs, parasites (*Mammomonogamus*)

CRITICAL DANGERS: 1. STERNAL RECUMBENCY = FATAL — abdominal organs crush diaphragm → respiratory arrest within minutes. ALWAYS lateral recumbency. 2. NO THORACOCENTESIS — no pleural cavity to drain; fluid in connective tissue is a clinical emergency. 3. ANAESTHESIA: IPPV parameters differ from other large mammals — standard equine/bovine protocols NOT applicable. 4. NECROPSY: Firm lung-chest adhesion is NORMAL — do NOT diagnose as fibrinous pleuritis.

6. Skull Morphology & Sexual Dimorphism

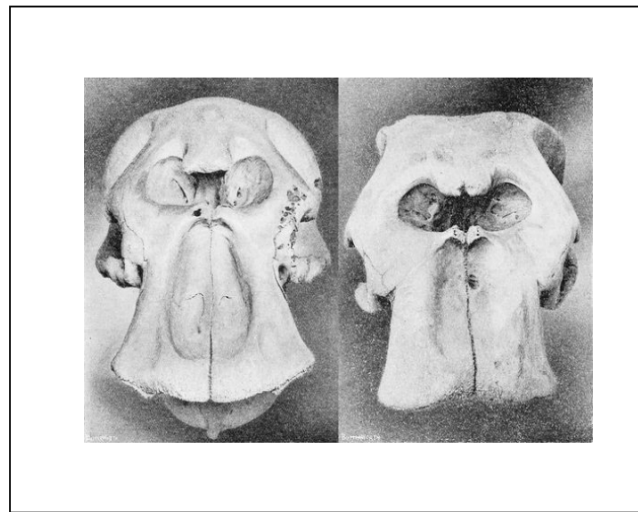


11 Slide 11 — Skull: Male vs Female (Frontal view) with labelled anatomy



12

Slide 12 — Skull: Lateral view, mandible comparison and photograph



12

Slide 13 — Skull photographs: male (left) vs female (right)

6.1 Skull Architecture

The elephant skull's imposing size is largely due to massive DIPLOIC SINUSES (honeycomb air spaces) in the frontal and parietal bones. This lightweight structure supports trunk and tusk weight without overloading the neck. The actual brain case is SMALL and positioned posteriorly.

- Incisive fossa (premaxilla): deep socket housing 1/3 of tusk length underground
- Temporal fossa: large — accommodates powerful temporal mastication muscles
- Bones present: interparietal, parietal, frontal, nasal, lacrimal, maxilla, premaxilla, temporal, occipital, mandible

6.2 Sexual Dimorphism — Frontal View

Feature	Label	Male Skull	Female Skull
Parieto-occipital crest	A	CONCAVE / depressed on median dorsal border — "saddle"	Rounded, CONVEX on dorso-median region — "dome"
Frontal bone	B	NARROW; "pinched" above nares; steep lateral walls	WIDER; more gradual lateral slope
External nares	C	HIGH above orbit; lateral edge slopes sharply down	Dumb-bell shape; less elevated
Incisive fossa	D	DEEP, narrow, "scooped out" below inferior nares border	Slopes gently; less abrupt
Tusk alveoli	E	LARGE, stout sockets; well-developed tusks	Small or absent alveoli; tushes or none

6.3 Sexual Dimorphism — Lateral View & Mandible

Feature	Label	Male	Female
Frontal (profile)	F	Slightly CONCAVE	Gently ROUNDED
Temporal line	G	PROMINENT muscle ridges	Smooth; minimal ridges
Premaxillaries	H	LARGE	Smaller
Parieto-occipital bosses	I	Large, SWOLLEN — prominent in older males	Absent or minimal
Mandible symphysis (chin)	--	ELONGATED, drops forward, THICK & BROAD	SHORT, THIN and POINTED

6.4 Quantitative Ratios

Ratio	Male	Female
Occipital length : Forehead length	~1.5–1.65 (greater)	~1.0 (almost equal)
Forehead length : Total head length	<0.5 (forehead <50% of head)	> 0.5 (forehead >50% of head)

- Dimorphism is AGE-DEPENDENT: <2 years skulls look similar in both sexes
- Full male features by age 25–30 years; tusk alveoli enlarge, bosses develop from age 10+
- Pelvic bone shows GREATER sexual dimorphism than skull (Loo et al. 2024, PMC12383435)
- Makhna (tuskless bulls): all other male skull features present EXCEPT tusk alveoli

6.5 Mortality Investigation — Skull-Based Sex Determination

Step	Action	Result
1	Check parieto-occipital crest (top rear)	Male: saddle; Female: dome
2	Assess frontal width above nares	Male: narrow/pinched; Female: wider
3	Examine incisive fossa depth	Male: deep/scooped; Female: gentle slope
4	Check tusk alveoli size	Male: large; Female: small/absent
5	Examine mandible chin	Male: elongated/thick; Female: short/pointed
6	Measure forehead:head ratio	Male: <0.5; Female: >0.5
7	Record ALL findings in mortality form	Document with photographs + scale bar

SKULL KEY POINTS: Definitive sex determination from skull alone even from fully skeletonised carcass. Large skull = mostly AIR SINUSES, not brain. Document tusk alveoli depth even if tusks removed by poachers.

7. Age Estimation — Molar Progression

Assessing age



Molar	No. of plates	Appearance age	Replacement age
I	4	4 months	2-2.5 yrs
II	8	6 months	6yrs
III	12	3 yrs	12 yrs
IV	12 (more wide)	8 yrs	25yrs
V	16	20yrs	50-60 yrs
VI	24	40yrs	Life long

14

Slide 14 — Molar table for assessing age with skull cross-section

7.1 Horizontal Molar Replacement — Unique Proboscidean Feature

Elephant molars develop at the BACK of the jaw and move FORWARD as each preceding molar wears down. Worn fragments drop off from the front. Six molar sets (M1–M6), each larger and longer-lasting than the last. Maximum 2 molars in use simultaneously on any one side. Final molar (M6) is never replaced.

7.2 Molar Identification & Age Table (Asian Elephant)

Molar	Enamel Plates (Lamellae)	Appearance Age	Worn Out By	Active Use Age Range	Notes
M1	4	4 months	2–2.5 yrs	0–2.5 yrs	Deciduous; very small; first to drop off
M2	8	6 months	~6 yrs	6 mths–6 yrs	Milk tooth; overlaps M1
M3	12	3 yrs	~12 yrs	3–12 yrs	First permanent-like molar
M4	12 (WIDER plates)	8 yrs	~25 yrs	8–25 yrs	SAME count as M3 but WIDER lamellae — key distinction
M5	16	20 yrs	50–60 yrs	20–60 yrs	Longest in use; present during prime adulthood
M6	24	~40 yrs	Death — never replaced	40 yrs–end of life	Final molar; exhaustion → starvation → death

7.3 Asian vs African Elephant Molar Timeline

Molar	Asian Elephant — Worn by	African Elephant — Worn by
M5	50–60 years	~40 years
M6 (last molar)	~65–70 years (life long)	~47 years
Max natural lifespan	65–70 years	~47–50 years

Death mechanism	M6 exhaustion → cannot chew → starvation	Same mechanism; earlier in African
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7.4 Necropsy Protocol — Molar Examination

Step	Action	Interpretation
1	Open jaw; expose molars on both sides	Identify active molar — flat occlusal surface = currently in use
2	Count enamel plates (lamellae) with brush/probe	M1=4, M2=8, M3=12, M4=12wide, M5=16, M6=24
3	Distinguish M3 vs M4 (both have 12 plates)	M4 lamellae are WIDER — measure width; a key distinction
4	Assess wear gradient (front-to-back tilt)	Anterior plates more worn = tooth in later stage → age at upper end of that molar range
5	Check for erupting molar behind active molar	Present = transitioning to next set → near upper age of current molar
6	Photograph with scale bar before removal	Mandatory for mortality investigation records
7	Record age estimate with range	E.g., "M5 in wear, ~60% worn → estimated age 35–50 years"

7.5 Additional Age Indicators

Feature	Young (<15 yrs)	Middle-aged (15–40 yrs)	Old (>45 yrs)
Temporal hollowing	None — full temples	Slight	Deep depression — "sunken temples"
Tusk size (male)	Short <50 cm	Medium 50–100 cm	Long >100 cm; often broken or worn
Skin texture	Relatively smooth	Progressively wrinkled	Deeply wrinkled; very dry
Skull sexual dimorphism	Minimal	Developing	Fully expressed; bosses prominent
Body condition	Well-muscled	Good	Often thin; skull/hip bones prominent

AGE ESTIMATION RULES: 1. ALWAYS examine molars — most reliable single indicator 2. NEVER age by tusk size alone — growth varies between individuals 3. Count lamellae with probe AFTER cleaning — do not estimate 4. Distinguish M3 vs M4 by lamellae WIDTH (not count) 5. M6 fully worn/absent → animal >60 years; death may be starvation from dental exhaustion 6. Document both sides (left/right, upper/lower) of the jaw 7. Give age as a RANGE, not a single number: "M5 in wear, ~50–60% worn → est. age 30–45 years"